

task inputs, routine and nonroutine, that are used to produce output Q , which sells at price one. Because our discussion stresses that computers neither strongly substitute nor strongly complement nonroutine manual tasks, we consider this model to pertain primarily to routine cognitive and routine manual tasks, and nonroutine analytic and nonroutine interactive tasks.

We assume for tractability an aggregate, constant returns to scale Cobb-Douglas production function of the form,

$$(1) \quad Q = (L_R + C)^{1-\beta} L_N^\beta, \quad \beta \in (0, 1),$$

where L_R and L_N are routine and nonroutine labor inputs and C is computer capital, all measured in efficiency units. Computer capital is supplied perfectly elastically at market price ρ per efficiency unit, where ρ is falling exogenously with time due to technical advances. The declining price of computer capital is the causal force in our model.⁸

We assume that computer capital and labor are perfect substitutes in carrying out routine tasks. Cobb-Douglas technology further implies that the elasticity of substitution between routine and nonroutine tasks is one, and hence computer capital and nonroutine task inputs are relative complements. While the assumption of perfect substitutability between computer capital and routine task input places assumptions A1 and A2 in bold relief, the only substantive requirement for our model is that computer capital is more substitutable for routine than nonroutine tasks. Observe that routine and nonroutine tasks are q -complements; the marginal productivity of nonroutine tasks rises with the quantity of routine task input, consistent with assumption A3.⁹

We assume a large number of income-maximizing workers, each of whom inelastically supplies one unit of labor. Workers have heterogeneous productivity endowments in both routine and nonroutine tasks, with $E_i = [r_i, n_i]$ and $1 \geq r_i, n_i > 0 \quad \forall i$. A given worker can choose to supply r_i efficiency units of routine task input, n_i efficiency units of nonroutine task input, or any convex

8. Borghans and ter Weel [2002] offer a related model exploring how the declining price of computer capital affects the diffusion of computers and the distribution of wages. A key difference is that the tasks performed by computers and workers are inseparable in the Borghans-ter Weel model. Accordingly, computerization alters wage levels but does not directly change the allocation of human labor input across task types. This latter point is the focus of our model and empirical analysis.

9. Specifically, $\partial^2 Q / \partial L_N \partial (L_R + C) = \beta(1 - \beta) L_N^{\beta-1} / (L_R + C)^\beta > 0$.

combination of the two. Hence, $L_i = [\lambda_i r_i, (1 - \lambda_i)n_i]$, where $0 \leq \lambda_i \leq 1$. These assumptions imply that workers will choose tasks according to comparative advantage as in Roy [1951]. We adopt the Roy framework because it implies that relative task supply will respond elastically to relative wage levels. If, instead, workers were bound to given tasks, the implications of our model for task productivity would be unchanged, but technical progress, reflected by a decline in ρ , would not generate re-sorting of workers across jobs.

Two main conditions govern market equilibrium in this model. First, given the perfect substitutability of routine tasks and computer capital, the wage per efficiency unit of routine task input is pinned down by the price of computer capital:¹⁰

$$(2) \quad w_R = \rho.$$

Second, worker self-selection among occupations—routine versus nonroutine—clears the labor market.

Define the relative efficiency of individual i at nonroutine versus routine tasks as $\eta_i = n_i / r_i$. Our assumptions above imply that $\eta_i \in (0, \infty)$. At the labor market equilibrium, the marginal worker with relative efficiency units η^* is indifferent between performing routine and nonroutine tasks when

$$(3) \quad \eta^* = w_R / w_N.$$

Individual i supplies routine labor ($\lambda_i = 1$) if $\eta_i < \eta^*$, and supplies nonroutine labor otherwise ($\lambda_i = 0$).

To quantify labor supply, write the functions $g(\eta)$, $h(\eta)$, which sum population endowments in efficiency units of routine and nonroutine tasks, respectively, at each value of η . Hence, $g(\eta) = \sum_i r_i \cdot I[\eta_i < \eta]$ and $h(\eta) = \sum_i n_i \cdot I[\eta_i \geq \eta]$, where $I[\cdot]$ is the indicator function. We further assume that η_i has nonzero support at all $\eta_i \in (0, \infty)$, so that that $h(\eta)$ is continuously upward sloping in η , and $g(\eta)$ is continuously downward sloping.

Assuming that the economy operates on the demand curve, productive efficiency requires

$$(4) \quad w_R = \frac{\partial Q}{\partial L_R} = (1 - \beta)\theta^{-\beta} \quad \text{and} \quad w_N = \frac{\partial Q}{\partial L_N} = \beta\theta^{1-\beta},$$

10. We implicitly assume that the shadow price of nonroutine tasks absent computer capital exceeds ρ and hence equation (2) holds with equality. In the precomputer era it is likely that $w_R < \rho$.

where θ in this expression is the ratio of routine to nonroutine task input in production:

$$(5) \quad \theta \equiv (C + g(\eta^*)) / h(\eta^*).$$

These equations provide equilibrium conditions for the model's five endogenous variables $(w_R, w_N, \theta, C, \eta)$. We use them to analyze how a decline in the price of computer capital affects task input, wages, and labor supply, beginning with the wage paid to routine task input.

It is immediate from (2) that a decline in the price of computer capital reduces w_R one-for-one, $\partial(\ln w_R) / \partial(\ln \rho) = 1$, and hence demand for routine task input rises:

$$(6) \quad \frac{\partial \ln \theta}{\partial \ln \rho} = -\frac{1}{\beta}.$$

From the perspective of producers, the rise in routine task demand could be met by either an increase in C or an increase in L_R (or both). Only the first of these will occur, however. Because routine and nonroutine tasks are productive complements (specifically q -complements), the relative wage paid to nonroutine tasks rises as ρ declines:

$$(7) \quad \frac{\partial \ln(w_N/w_R)}{\partial \ln \rho} = -\frac{1}{\beta} \quad \text{and} \quad \frac{\partial \ln \eta^*}{\partial \ln \rho} = \frac{1}{\beta}.$$

Consequently, marginal workers will reallocate their labor input from routine to nonroutine tasks. Increased demand for routine tasks must be met entirely by an influx of computer capital.

Hence, an exogenous decline in the price of computer capital raises the marginal productivity of nonroutine tasks, causing workers to reallocate labor supply from routine to nonroutine task input. Although routine labor input declines, an inflow of computer capital more than compensates, yielding a net increase in the intensity of routine task input in production.

I.B. Industry Level Implications

Does this model have testable microeconomic implications? Because the causal force in the model is the price of computer capital, in one sense we have only a single macroeconomic time series available. However, additional leverage may be gained by

considering equation (1) as representing the production function of a single industry, with distinct industries j producing outputs q_j that demand different mixes of routine and nonroutine tasks. We write industry j 's production function as

$$(8) \quad q_j = r_j^{1-\beta_j} n_j^{\beta_j}, \quad \beta_j \in (0, 1),$$

where β_j is the industry-specific factor share of nonroutine tasks, and r_j , n_j denote the industry's task inputs. All industries use Cobb-Douglas technology, but industries with smaller β_j are more routine task intensive.

We assume that consumer preferences in this economy may be represented with a Dixit-Stiglitz [1977] utility function,

$$(9) \quad U(q_1, q_2, \dots, q_j) = \left(\sum_j q_j^{1-v} \right)^{1/(1-v)},$$

where $0 < v < 1$. The elasticity of demand for each good is $-(1/v)$, with the market clearing price inversely proportional to the quantity produced, $p_j(q_j) \propto q_j^{-v}$.

Industry profit maximization yields the following first-order conditions for wages:

$$(10) \quad \begin{aligned} \rho &= n_j^{\beta_j} r_j^{-\beta_j} (1 - \beta_j) (1 - v) (n_j^{\beta_j} r_j^{1-\beta_j})^{-v} \quad \text{and} \\ w_N &= n^{\beta_j-1} r^{1-\beta_j} \beta_j (1 - v) (n_j^{\beta_j} r_j^{1-\beta_j})^{-v}. \end{aligned}$$

Rearranging to obtain factor demands gives

$$(11) \quad \begin{aligned} n_j &= w_N^{-1/v} (\beta_j (1 - v))^{1/v} \left(\frac{w_N}{\rho} \cdot \frac{(1 - \beta_j)}{\beta_j} \right)^{((1-\beta_j)(1-v))/v} \quad \text{and} \\ r_j &= \rho^{-1/v} ((1 - \beta_j)(1 - v))^{1/v} \left(\frac{w_N}{\rho} \cdot \frac{(1 - \beta_j)}{\beta_j} \right)^{(\beta_j(v-1))/v}. \end{aligned}$$

Using these equations, we obtain the following three propositions, which we test empirically below.

- P1. Although all industries face the same price of computer capital, ρ , the degree to which industries adopt this capital as its price declines depends upon β_j . For a given price decline, the proportionate increase in demand for routine task input is larger in routine-task-intensive (β_j small) industries, as may be seen by taking the cross-partial derivative of routine task demand with respect to ρ and β_j :